

Appendix S7: Case Study Preliminary Analysis Summary Report

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1 Preliminary Analysis of VEFMAP Riparian Vegetation Surveys

A pilot study was conducted to identify model structures that were estimable (i.e., generated reliable parameter estimates) and captured key features of the data (e.g., many zeros). The details are outlined in the preregistration document. In short, the process began with a preferred model structure and refined this by removing interactions until a suitable model was identified.

Listing 1.1 The preferred full model structure (in high-level, R notation).

```
Veg. response ~ wpfg * origin * flow predictors +  
wpfg * origin * zone * period +  
grazing +  
(1 | site / transect) +  
(1 | transect / metres) +  
(1 | survey_year) +  
(1 | species) +  
offset(npoints)
```

The model for vegetation cover included an additional term for the previous cover in the previous survey, which made this an autoregressive model (i.e., modelling the change in cover rather than absolute cover).

The flow predictors were simple metrics intended to capture broad features of the flow regime at a site. The two metrics were the number of days in which discharge was above baseflow levels, and the number of days in which discharge was above spring fresh levels. Values were calculated for each water year, and a 3-year rolling mean was used to represent broad changes in the multi-year flow regime (cf. variation within or among individual years). Quadratic transforms of these metrics were also included in the model.

Models for cover and species richness were fitted with zero-inflated Poisson distributions, with negative binomial distributions as an alternative to address over-dispersion. For cover, this differs from the (generally preferred) use of a bounded distribution (e.g., binomial or beta). However, parameters of bounded distributions are challenging to estimate when many values are near the bounds (lower or upper), and typically introduce link functions that do not allow straightforward specifications of autoregressive models. For this reason, the Poisson distribution was used in models of cover, noting that this approximates the binomial distribution in the limit of large numbers of trials.

Initial models aimed to include data at the level of species. This was infeasible for two reasons. First, data resolved to the level of species required an extremely large data set (due to the many species recorded), which made model estimation computationally challenging. Second, the large data set (with all species) comprised mostly zero observations, resulting in extreme zero-inflation that affected model convergence. The final models were fitted to cover summed over all species within a plant functional grouping, noting that these summed cover estimates sometimes (but rarely) exceeded 100% due to species overlap.

1.1 Key decisions and final model structures

1. High-level interactions are too challenging to estimate. Instead proposed three models for vegetation cover:
 1. A flow regime model that examines flow predictors only (with random effects).
 2. A flow event model (version 1) that examines **zone-by-period** interactions but assumes that all functional groups have similar patterns (within a **zone** and **period**).
 3. A flow event model (version 2) that examines functional group-by-period interactions but assumes that all zones respond similarly.
2. Species richness model is still being finalised, but should at least be able to encompass versions 1 and 2 of the cover model in a single analysis.
3. These models ask high-level questions about the effects of flows (and flow events) on vegetation cover and species richness and provide generalisable insight into how flows could be delivered to influence vegetation outcomes.
4. Quadratic flow effects too complex to fit: removed.
5. Metres within transect too complex to include: replaced with a random effect for metres, which assumes that the distance up the bank might influence vegetation outcomes but in similar ways among sites.
6. Zero-inflation specified as a function of plant functional group to allow for different proportions of zeros among groupings.
7. Over-dispersion present but not resolved by negative binomial model (even with **wpfg**-specific dispersion parameters): zero-inflated Poisson model retained, acknowledging lingering issues of over-dispersion.
8. Offset removed because all observations were based on the same number of points.

1.2 Decision points for full analysis

Several decisions will need to be made prior to the full analysis (which should begin ASAP).

1. Assess whether the full **zone-by-period-by-wpfg** interaction is possible with the full data set. If not, use simplified models outlined above.
2. Decide whether to fit a single model to all waterbodies or separate models for each (decision based on model convergence and posterior checks – a single model is preferable if feasible).
3. Decide on how to visualise model outputs.

2 Vegetation Cover: regime model

This model includes two flow predictors (days above baseflow, days above spring fresh) and the full suite of random effects. The aim is to examine how past flow conditions influence vegetation cover. The model does not consider `zone` or `period`.

The model identifies functional group-specific impacts of days above baseflow or spring fresh levels. Interpretations here are based on the predicted associations (Figure 2.2 and Figure 2.3) because interpreting forest plots of coefficients (Figure 2.1) is challenging in models with many interactions. Positive effects of baseflows are clearest for native `ATl` and `ATe` and exotic `Tdr` and `Tda`, all of which have higher cover under baseflows. Positive effects of spring freshes are clearest for native `Sk`, `Tda`, and `Ate` and exotic `ATe`. Negative effects of spring freshes are clearest for `Tdr`.

Listing 2.1 Vegetation cover, regime model structure.

```
cover_ar_TMBmod_1 <- glmmTMB::glmmTMB(  
  hits ~ log_hits_tml +  
  days_above_baseflow_std * wpfg * origin +  
  days_above_springfresh_std * wpfg * origin +  
  # days_above_baseflow_std^2 +  
  # days_above_springfresh_std^2 +  
  # zone * period +  
  # zone + period +  
  # grazing + wpfg +  
  (1 | site / transect) +  
  # (1 | site / period) +  
  (1 | metres) +  
  (1 | survey_year),  
  # offset(npoint),  
  family = poisson,  
  ziformula = ~ wpfg,  
  # dispformula = ~ wpfg ,  
  data = veg_cover_ar_sum |>  
    filter(!wpfg_ori %in% c("Atl_native",  
                           "Ate_native",  
                           "Tda_unknown"))  
)
```

Table 2.1: Vegetation cover flow regime model summary. Signif. codes: 0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1.

Conditional model	Estimate	z value	Std. Error	Pr($\geq z $)
(Intercept)	-6.338926	3.945021	-1.61	0.1081
log_hits_tml	0.466865	0.084717	98.98	< 2e-16 ***
days_above_baseflow_std	3.208391	3.369341	0.95	0.3410
wpgATe	6.158428	3.941779	1.56	0.1182
wpfgATl	3.547942	0.684107		5.19 2.15e-07 ***
wpfgSe	-21.385108	25.954264	-0.82	0.4100
wpfgsk	1.108790	1.345171	0.82	0.4132
wpfgTda	7.307450	3.941643	1.85	0.0638
wp fgTdr	7.736796	3.941596	1.96	0.0497 *
originative	2.784402	3.994830	0.70	0.4858
days_above_springfresh_std	-0.123400	0.390857	-0.32	0.7522
days_above_baseflow_std:wpfgATe	-2.987176	3.367936	-0.89	0.3751
days_above_baseflow_std:wpfgATl	0.155333	0.657344	0.24	0.8132
days_above_baseflow_std:wpfgSe	14.468021	26.401090	0.55	0.5837
days_above_baseflow_std:wpfgSk	0.182377	1.265110	0.14	0.8854
days_above_baseflow_std:wpfgTda	-2.975603	3.367609	-0.88	0.3769
days_above_baseflow_std:wpfgTdr	-2.989015	3.367573	-0.89	0.3748
days_above_baseflow_std:originative	-2.304494	3.430569	-0.67	0.5017
wpfgAle: originative	-1. 514880	3.995046	-0.38	0.7045
wpfgATl:originative	NA	NA	NA	NA
wpfgSe: originative	NA	NA	NA	NA
wpfgSk:originative	NA	NA	NA	NA
wpfgida: originative	-2.579748	3.994900	-0.65	0.5184
wpfgTdr: originative	-3.710273	3.995021	-0.93	0.3530
wpfgATe:days_above_springfresh_std	1.010669	0.382893	2.64	0.0083 **
wpfgATl:days_above_springfresh_std	-1.755976	0.272420		-6.45 1.15e-10 ***
wpfgSe:days_above_springfresh_std	2.328011	3.084072	0.75	0.4503
wpfgSk:days_above_springfresh_std	-0.323750	0.546548	-0.59	0.5536
wpfgTda:days_above_springfresh_std	0.255564	0.381392	0.67	0.5028
wpfgTdr:days_above_springfresh_std	0.271463	0.381315	0.71	0.4765
originative:days_above_springfresh_std	2.036161	0.468210		4.35 1.37e-05 ***
days_above_baseflow_std:wpfgATe:originative	2.376384	3.430972	0.69	0.4885
days_above_baseflow_std:wpfgATl:originative	NA	NA	NA	NA

Table 2.1: Vegetation cover flow regime model summary. Signif. codes: 0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1.

Conditional model	Estimate	z value	Std. Error	Pr($\geq z $)
days_above_baseflow_std:wpfgSe:originnative	NA	NA	NA	NA
days_above_baseflow_std:wpfg5k:originnative	NA	NA	NA	NA
days_above_baseflow_std:wpfgTda:originnative	2.166421	3.430644	0.63	0.5277
days_above_baseflow_std:wpfgTdr:originnative	2.291317	3.430663	0.67	0.5042
wpfgATe:originnative:days_above_springfresh_std	-2.761693	0.469796	-5.88	4.14e-09 ***
wpfgATl:originnative:days_above_springfresh_std	NA	NA	NA	NA
wpfgSe:originnative:days_above_springfresh_std	NA	NA	NA	NA
wpfgSk:originnative:days_above_springfresh_std	NA	NA	NA	NA
wpfgTda:originnative:days_above_springfresh_std	-1.953152	0.468932	-4.17	3.11e-05 ***
wpfgTdr:originnative:days_above_springfresh_std	-2.770192	0.472856	-5.86	4.67e-09 ***

2.1 Vegetation Cover: flow event model (version 1)

This model includes several categorical predictors (**origin**, **wpfg**, **grazing**, and a **zone-by-period** interaction) and the full suite of random effects. The aim is to examine how vegetation cover changes in specific zones before and after key flow events (spring and summer freshes). This first version allows functional groups to have different levels of cover *but assumes that changes following flow events are similar in all groups within a zone*. Although this is not an ideal model structure, it is likely that functional groups are restricted to particular zones, in which case the **zone-by-period** interaction may capture some of the variation attributable to functional groupings.

The model indicated that vegetation cover differed in its responses to spring and summer freshes in each zone (below baseflow, baseflow to spring fresh, above spring fresh). Below baseflow level, vegetation increased following the spring fresh and remained high following the summer event (Figure 2.5). In the baseflow-to-spring fresh zone, vegetation increased following the spring fresh but returned to pre-spring levels following the summer fresh (Figure 2.5). In the above-spring fresh zone, vegetation level declined following both the spring and the summer fresh (Figure 2.5).

Listing 2.2 Vegetation cover, flow event model (version 1).

```
cover_ar_TMBmod_2 <- glmmTMB::glmmTMB(
  hits ~ log_hits_tml +
  # days_above_baseflow_std * wpfg * origin +
  # days_above_springfresh_std * wpfg * origin +
  # days_above_baseflow_std^2 +
  # days_above_springfresh_std_sq +
  zone * period +
  origin + wpfg +
  grazing +
  (1 | site / transect) +
  # (1 | site / period) +
  (1 | metres) +
  (1 | survey_year),
  # offset(npoint),
  family = poisson,
  ziformula = ~ wpfg,
  # dispformula = ~ wpfg ,
  data = veg_cover_ar_sum |>
  filter(!wpfg_ori %in% c("Atl_native",
                          "Ate_native",
                          "Tda_unknown"))
)
```

Table 2.2: Vegetation cover flow event model (version 1) summary. Signif. codes: 0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1.

Conditional model	Estimate	z value	Std. Error	Pr($\geq z $)
(Intercept)	0.507407	0.202834	2.50	0.0124 *
log_hits_tml	0.503130	0.004658	108.00	< 2e-16 ***
zonebaseflow_to_springfresh	0.135690	0.019849	6.84	8.14e-12 ***
zonebelow_baseflow	0.027219	0.062187	0.44	0.6616

Conditional model	Estimate	z value	Std. Error	$\Pr(\geq z)$
periodafter_summer	-0.095429	0.019586	-4.87	1.10e-06 ***
periodbefore_spring	0.275100	0.019116	14.39	< 2e-16 ***
originnative	0.022715	0.011601	1.96	0.0502 .
wpfgATe	0.568841	0.074505	7.63	2.26e-14 ***
wpfgATl	0.392510	0.077440	5.07	4.01e-07 ***
wpfgSe	-0.532349	0.235144	-2.26	0.0236 *
wpfgSk	0.785098	0.158281	4.96	7.04e-07 ***
wpf_gTda	0.580179	0.075459	7.69	1.49e-14 ***
wpfgTdr	0.740681	0.076091	9.73	< 2e-16 ***
grazingY	-0.414008	0.207655	-1.99	0.0462 *
zonebaseflow_to_springfresh:periodafter_summer	-0.049931	0.022829	-2.19	0.0287 *
zonebelow_baseflow:periodafter_summer	0.072623	0.060393	1.20	0.2292
zonebaseflow_to_springfresh:periodbefore_spring	-0.474324	0.023176	-20.47	< 2e-16 ***
zonebelow_baseflow:periodbefore_spring	-0.844370	0.080058	-10.55	< 2e-16 ***

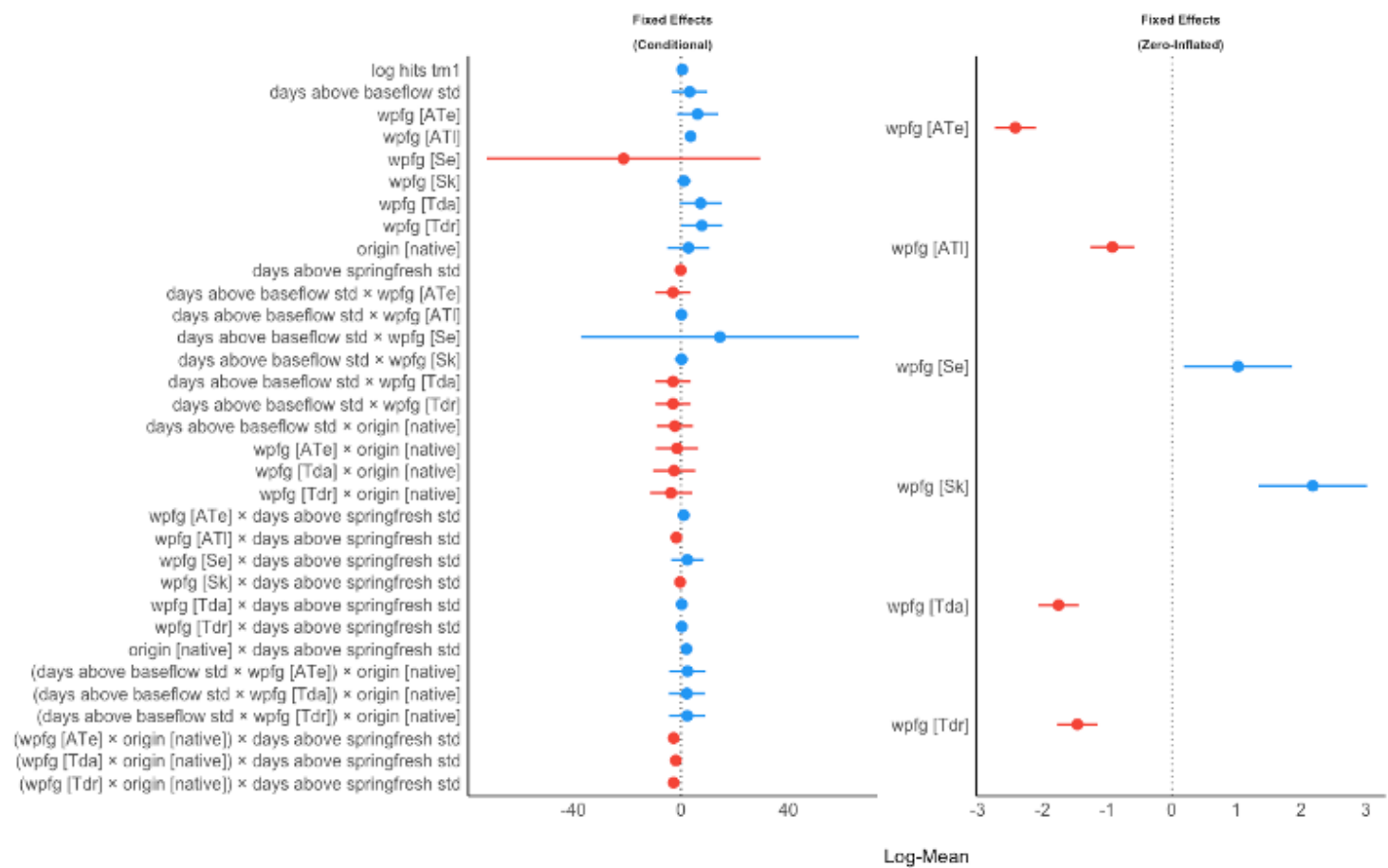


Figure 2.1: Effects plot for vegetation cover flow regime model.

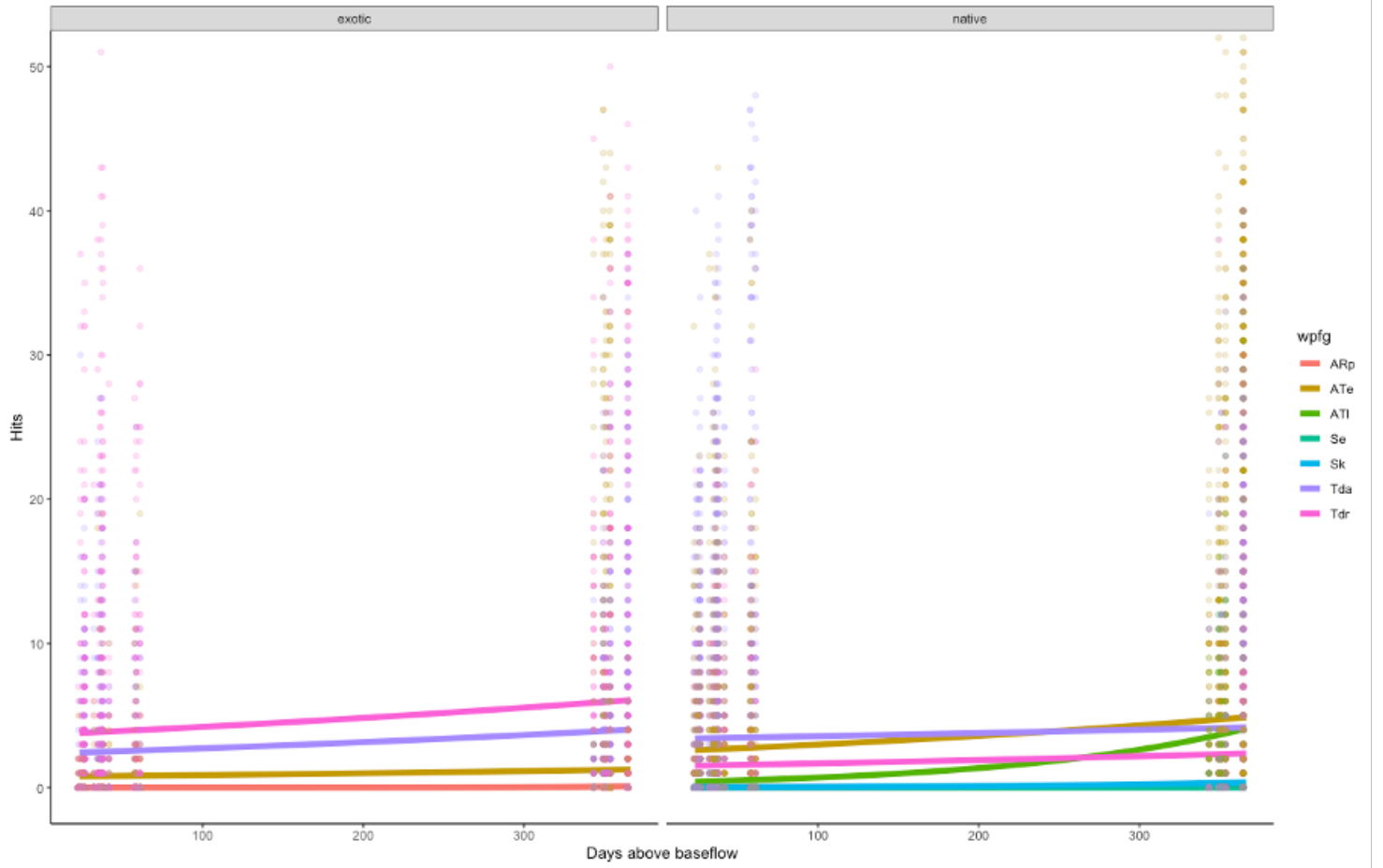


Figure 2.2: Model predicted plant cover for the three way interaction between Days above baseflow, functional group and origin. Points are raw data.

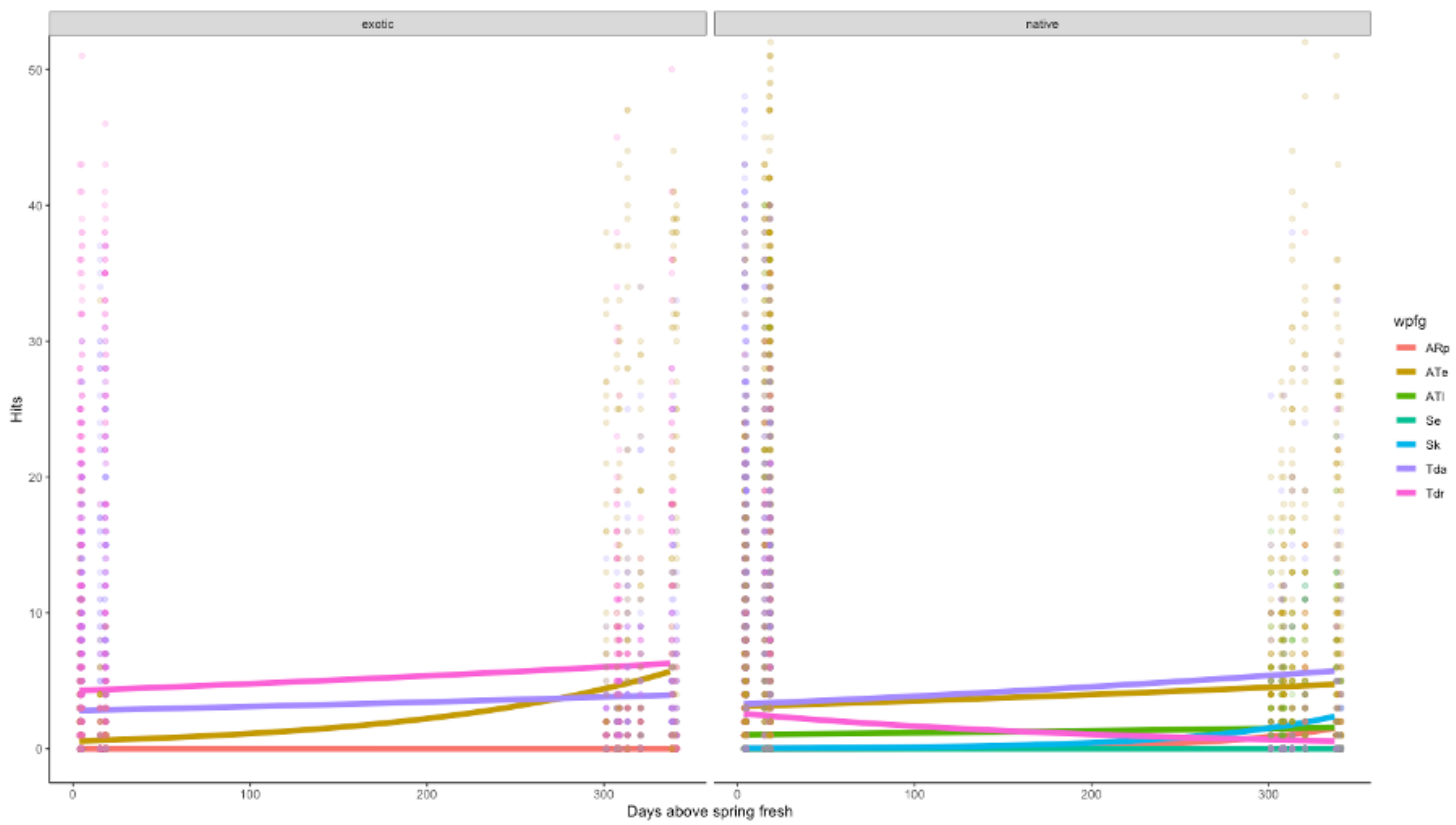


Figure 2.3: Model predicted plant cover for the three way interaction between Days above springfresh, functional group and origin. Points are raw data.

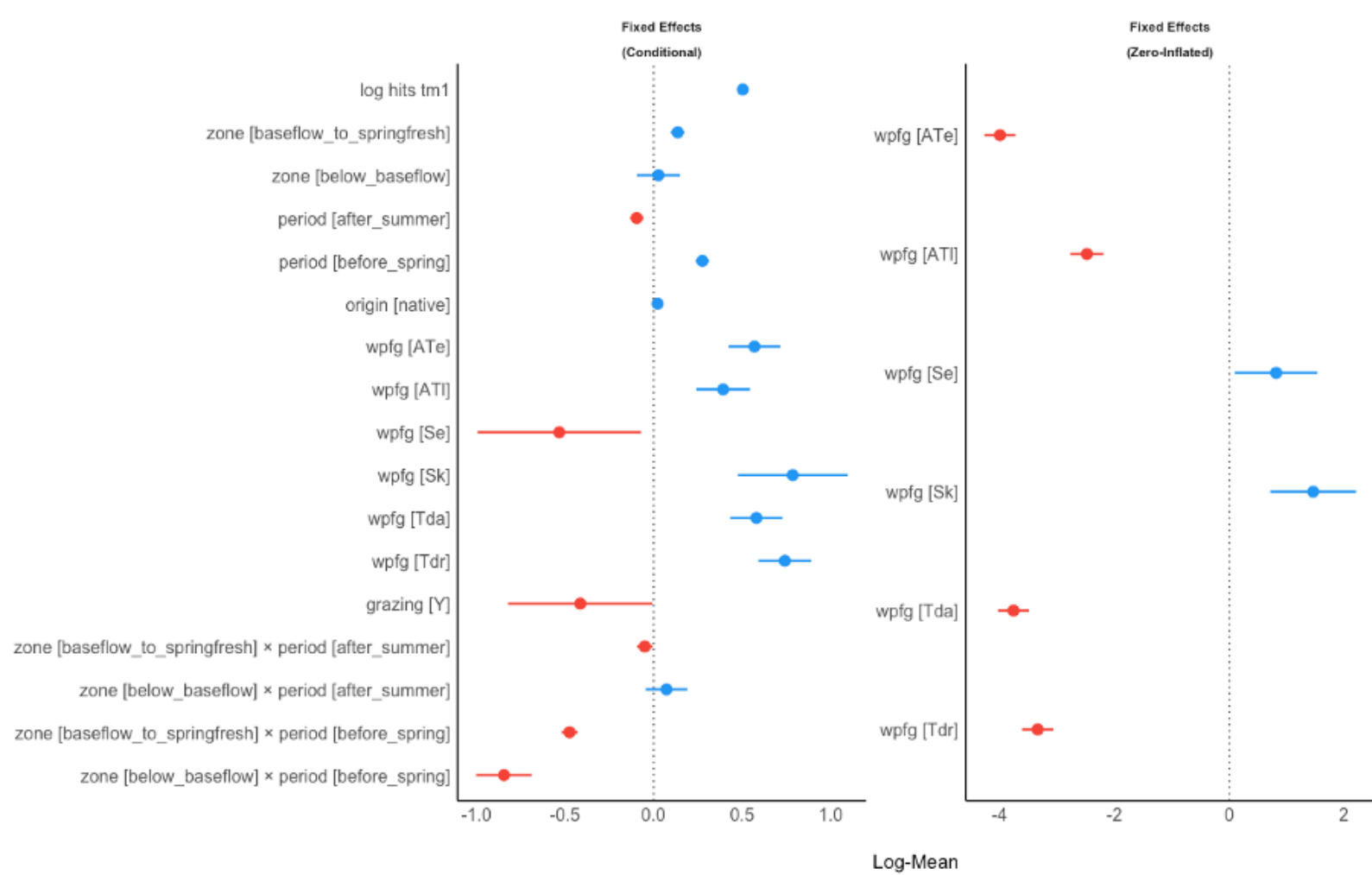


Figure 2.4: Vegetation cover flow event model (version 1) effects plot.

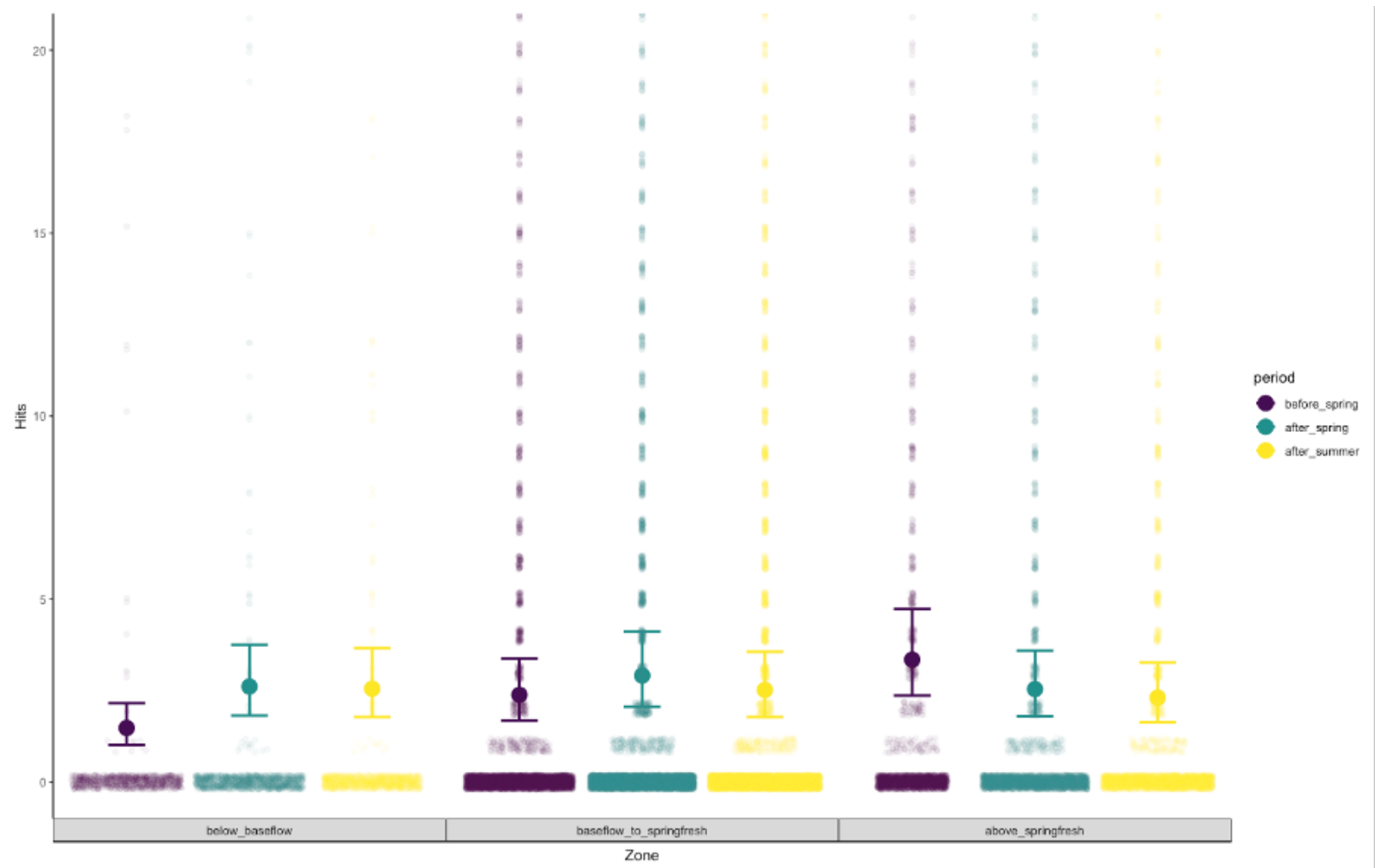


Figure 2.5: Model predicted plant cover for the two way interaction between **period** and **zone**. Points are raw data with the spread across x-axis representing data density.

2.2 Vegetation Cover: flow event model (version 2)

This model includes several categorical predictors (**origin**, **zone**, **grazing**, and a **wpfg**-by-**period** interaction) and the full suite of random effects. The aim is to examine how vegetation cover of each functional grouping changes before and after key flow events (spring and summer freshes). This second version allows functional groups to have different responses to flow events (spring and summer freshes) *but assumes that vegetation in all zones changes similarly following each flow event*. As for version 1, this is not an ideal model structure but provides an avenue to tease apart **wpfg**-specific responses to flow events.

The model indicated that cover of each functional group differed in its response to spring and summer freshes (Figure 2.7). Key patterns were as follows:

- **ARp**: increased after the spring fresh and again after the summer fresh.
- **ATe**: increased after the spring fresh and returned to pre-spring levels after the summer fresh.
- **ATl**: as for **ATe**.
- **Se**: increased after the spring fresh and largely maintained this level after the summer fresh.
- **Sk**: no change following the spring fresh but large increases following the summer fresh.
- **Tda**: no change following either flow event.
- **Tdr**: reduced following spring fresh then recovered slightly following the summer fresh, but not to pre-spring levels.

Listing 2.3 Vegetation cover, flow event model (version 2).

```
cover_ar_TMBmod_3 <- glmmTMB::glmmTMB(
  hits ~ log_hits_tml +
  # days_above_baseflow_std * wpfg * origin +
  # days_above_springfresh_std * wpfg * origin +
  # days_above_baseflow_std^2 +
  # days_above_springfresh_std_sq +
  zone + wpfg * period +
  origin +
  grazing +
  (1 | site / transect) +
  #(1 | site / period) +
  (1 | metres) +
  (1 | survey_year),
  # offset(npoint),
  family = poisson,
  ziformula = ~ wpfg,
  #dispformula = ~ wpfg ,
  data = veg_cover_ar_sum |>
    filter(!wpfg_ori %in% c("Atl_native",
                           "Ate_native",
                           "Tda_unknown")))
)
```

Table 2.3: Vegetation cover, flow event model (version 2) summary. Signif. codes: 0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1.

Conditional model	Estimate	z value	Std. Error	Pr($\geq z $)
(Intercept)	0.490933	0.220532	2.23	0.026006 *
log_hits_tml	0.526108	0.004773	110.22	< 2e-16 ***
zonebaseflow_to_springfresh	-0.020485	0.014807	-1.38	0.166522
zonebelow_baseflow	-0.135442	0.051703	-2.62	0.008803 **
wpfgATe	0.762082	0.120587	6.32	2.62e-10 ***
wpfgATL	0.826599	0.124319	6.65	2.95e-11 ***
wpfgSe	-0.219954	0.361766	-0.61	0.543187
wpfgSk	-0.409777	0.497890	-0.82	0.410493

Conditional model	Estimate	z value	Std. Error	$\Pr(\geq z)$
wpfgTda	0.640675	0.121310	5.28	1.28e-07 ***
wpfgTdr	0.361223	0.122625	2.95	0.003222 **
periodafter_summer	0.259131	0.152882	1.69	0.090080 .
periodbefore_spring	-0.769916	0.262410	-2.93	0.003346 **
originnative	0.022738	0.011705	1.94	0.052058 .
grazingY	-0.406325	0.203425	-2.00	0.045780 *
wpfgATe:periodafter_summer	-0.570432	0.153550	-3.71	0.000203 ***
wpfgATl:periodafter_summer	-0.864219	0.159835	-5.41	6.41e-08 ***
wpfgSe:periodafter_summer	-0.445005	0.462139	-0.96	0.335586
wpfgSk:periodafter_summer	1.502212	0.526754	2.85	0.004347 **
wpfgTda:periodafter_summer	-0.310229	0.153846	-2.02	0.043749
wpfgTdr: periodafter_summer	0.052045	0.155180	0.34	0.737333
wpfgATe:periodbefore_spring	0.414490	0.262945	1.58	0.114948
wpfgATl:periodbefore_spring	-0.060499	0.268355	-0.23	0.821633
wpfgSe:periodbefore_spring	-0.968411	0.865331	-1.12	0.263088
wpfgSk:periodbefore_spring	0.760442	0.736038	1.03	0.301531
wpfgTda:periodbefore_spring	0.611485	0.263133	2.32	0.020133 *
wpfgTdr:periodbefore_spring	1.539382	0.263473	5.84	5.14e-09 ***

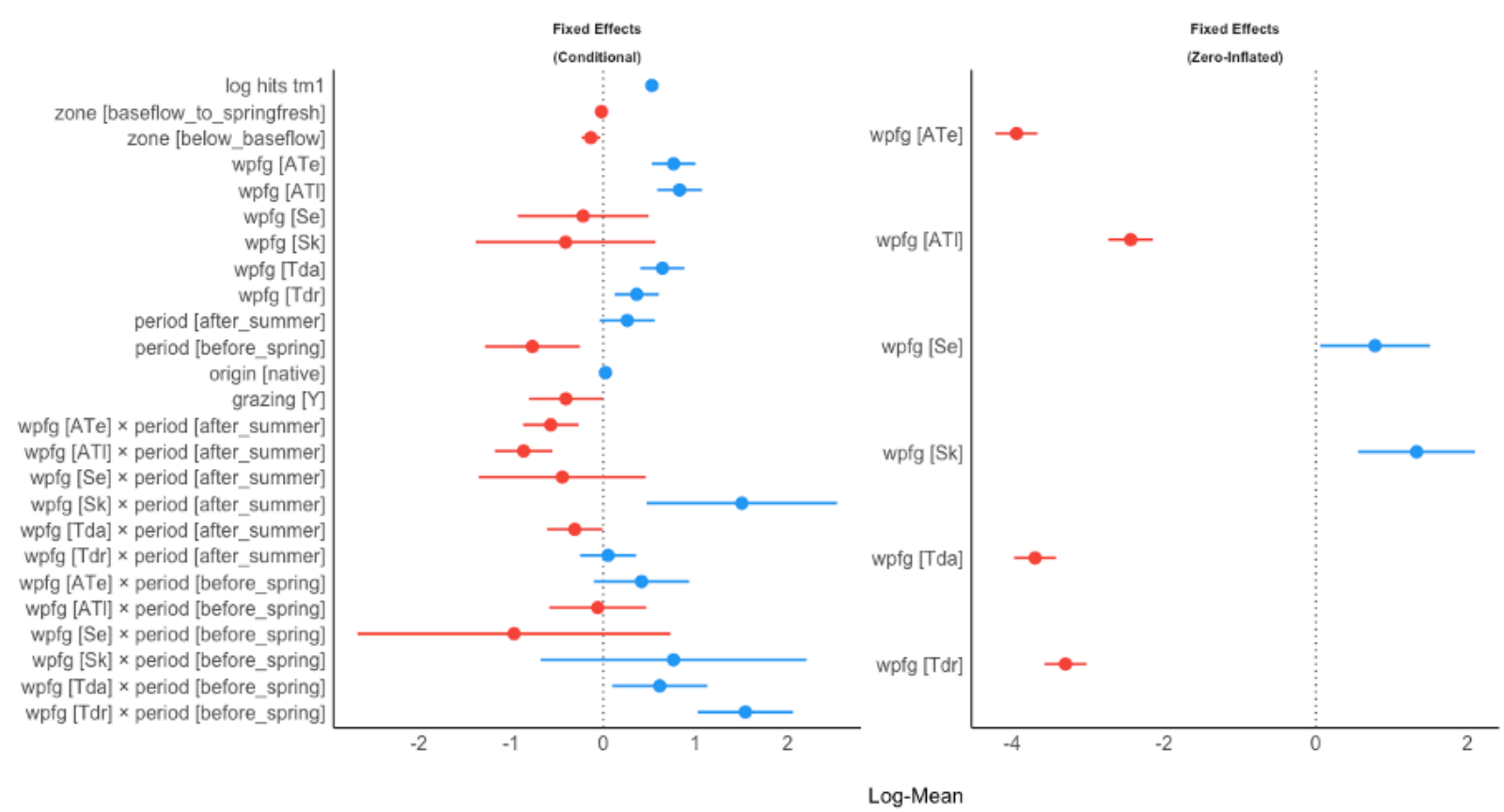


Figure 2.6: Vegetation cover, flow event model (version 2) effects plot.

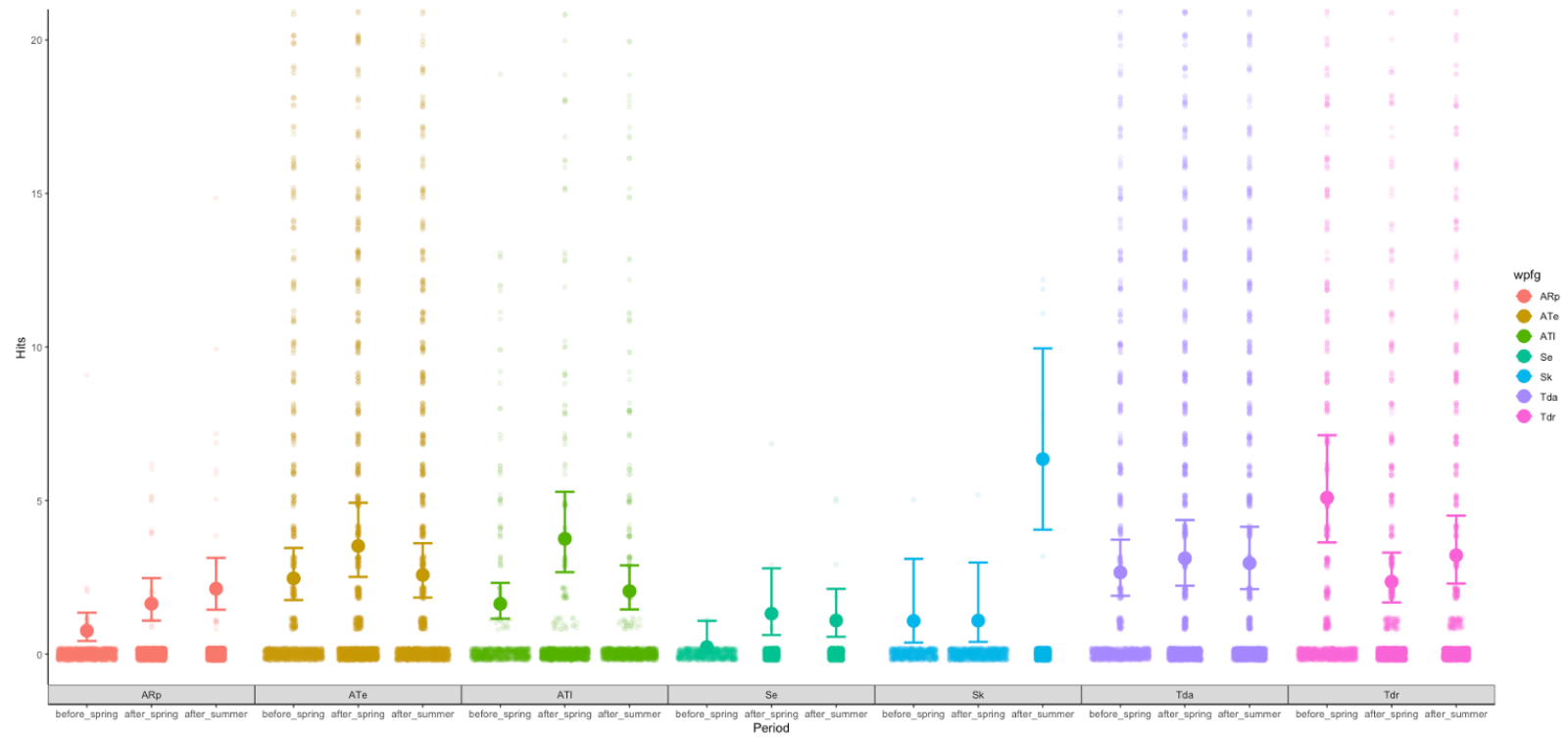


Figure 2.7: Model predicted plant cover for the two way interaction between **period** and **functional group**. Points are raw data with the spread across x-axis representing data density.

2.3 Species Richness Model

Species richness models are not fully determined yet. The below structure captures similar patterns to version 1 and version 2 of the cover model. The model includes two-way interactions between **zone** and **period**, **zone** and **plant functional group**, and **period** and **plant functional group**. Together, these terms aim to identify responses to flow events within a given zone or functional group, while accounting for differences in species richness of each functional group among zones.

The models illustrate a general pattern of relatively low temporal variation in species richness once accounting for spatial patterns due to zone and functional grouping. Figure 2.8 illustrates that total species richness does not change substantially within each zone following the spring or summer fresh. Similarly, Figure 2.10 shows that total species richness of each plant functional group does not change substantially following spring or summer freshes. By contrast, Figure 2.10 illustrates that the species richness of each plant functional group differs among zones. As expected, aquatic groupings are more species-richness below the baseflow level, whereas terrestrial groupings are more species richness above the spring fresh level.

JDY comment on Listing 2.4: HW to update, the `dispformula` is probably ignored for a Poisson model, use `ziformula` instead. Can flow metrics be included?

Listing 2.4 Species richness model structure.

```
richness_ar_TMBmod_1 <- glmmTMB::glmmTMB(
  richness ~
  days_above_baseflow_std * wpfg * origin +
  days_above_springfresh_std * wpfg * origin +
  # days_above_baseflow_std^2 + days_above_springfresh_std^2 +
  # zone * period +
  # zone * period + zone * wpfg + wpfg * period +
  # grazing + origin +
  (1 | site / transect) +
  # (1 | site / period) +
  (1 | metres) +
  (1 | survey_year),
  # offset(npoint),
  family = poisson,
  # family = nbinom2,
  # ziformula = ~ wpfg,
  # dispformula = ~ wpfg,
  data = veg_richness |>
    filter(!wpfg_ori %in% c("Atl_native",
                           "Ate_native",
                           "Tda_unknown"))
)
```

Best fitting so far – the three way interaction had convergence issues.

Table 2.4: Species richness model summary. Signif. codes: 0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1.

Conditional model	Estimate	z value	Std. Error	Pr(> z)
(Intercept)	0.490933	0.220532	2.23	0.026006 *
log_hits_tml	0.526108	0.004773	110.22	< 2e-16 ***
zonebase- flow_to_springfresh -0.020485		0.014807	-1.38	0.166522
zonebelow_baseflow	-0.135442	0.051703	-2.62	0.008803 **
wpfgATe	0.762082	0.120587	6.32	2.62e-10 ***
wpfgATL	0.826599	0.124319	6.65	2.95e-11
wpfgSe	-0.219954	0.361766	-0.61	0.543187
wpfgSk	-0.409777	0.497890	-0.82	0.410493
wpfgTda	0.640675	0.121310	5.28	1.28e-07 ***

Conditional model	Estimate	z value	Std. Error	Pr(> z)
wpfgTdr	0.361223	0.122625	2.95	0.003222 **
periodafter_summer	0.259131	0.152882	1.69	0.090080
periodbefore_spring	-0.769916	0.262410	-2.93	0.003346 **
originnative	0.022738	0.011705	1.94	0.052058
grazingY	-0.406325	0.203425	-2.00	0.045780 *
wpfgATe:	-0.570432	0.153550	-3.71	0.000203 ***
periodafter_summer				
wpfgATl:peri-	-0.864219	0.159835	-5.41	6.41e-08 *
odafter_summer				
wpfgSe:peri-	-0.445005	0.462139	-0.96	0.335586
odafter_summer				
wpfgSk:peri-	1.502212	0.526754	2.85	0.004347 **
odafter_summer				
wpfgTda:peri-	-0.310229	0.153846	-2.02	0.043749 *
odafter_summer				
wpfgTdr:peri-	0.052045	0.155180	0.34	0.737333
odafter_summer				
wpfgATe:periodbe-	0.414490	0.262945	1.58	0.114948
fore_spring				
wpfgATl:periodbe-	-0.060499	0.268355	-0.23	0.821633
fore_spring				
wpfgSe:periodbe-	-0.968411	0.865331	-1.12	0.263088
fore_spring				
wpfgSk:periodbe-	0.760442	0.736038	1.03	0.301531
fore_spring				
wpfgTda:periodbe-	0.611485	0.263133	2.32	0.020133 *
fore_spring				
wpfgTdr:periodbe-	1.539382	0.263473	5.84	5.14e-09 ***
fore_spring				

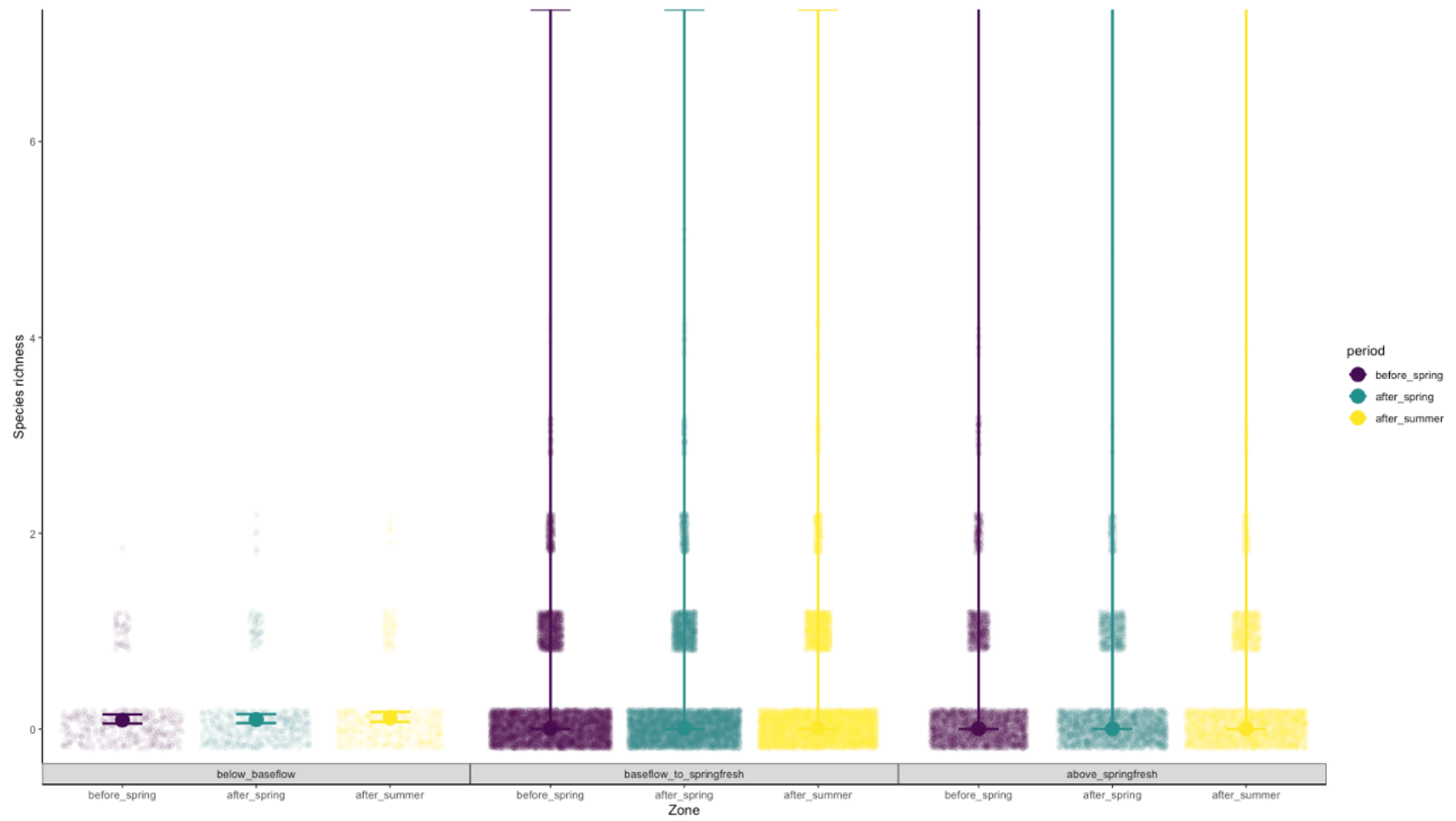


Figure 2.8: Model predicted species richness for the two way interaction between **period** and **zone**. Points are raw data with the spread across x-axis representing data density.

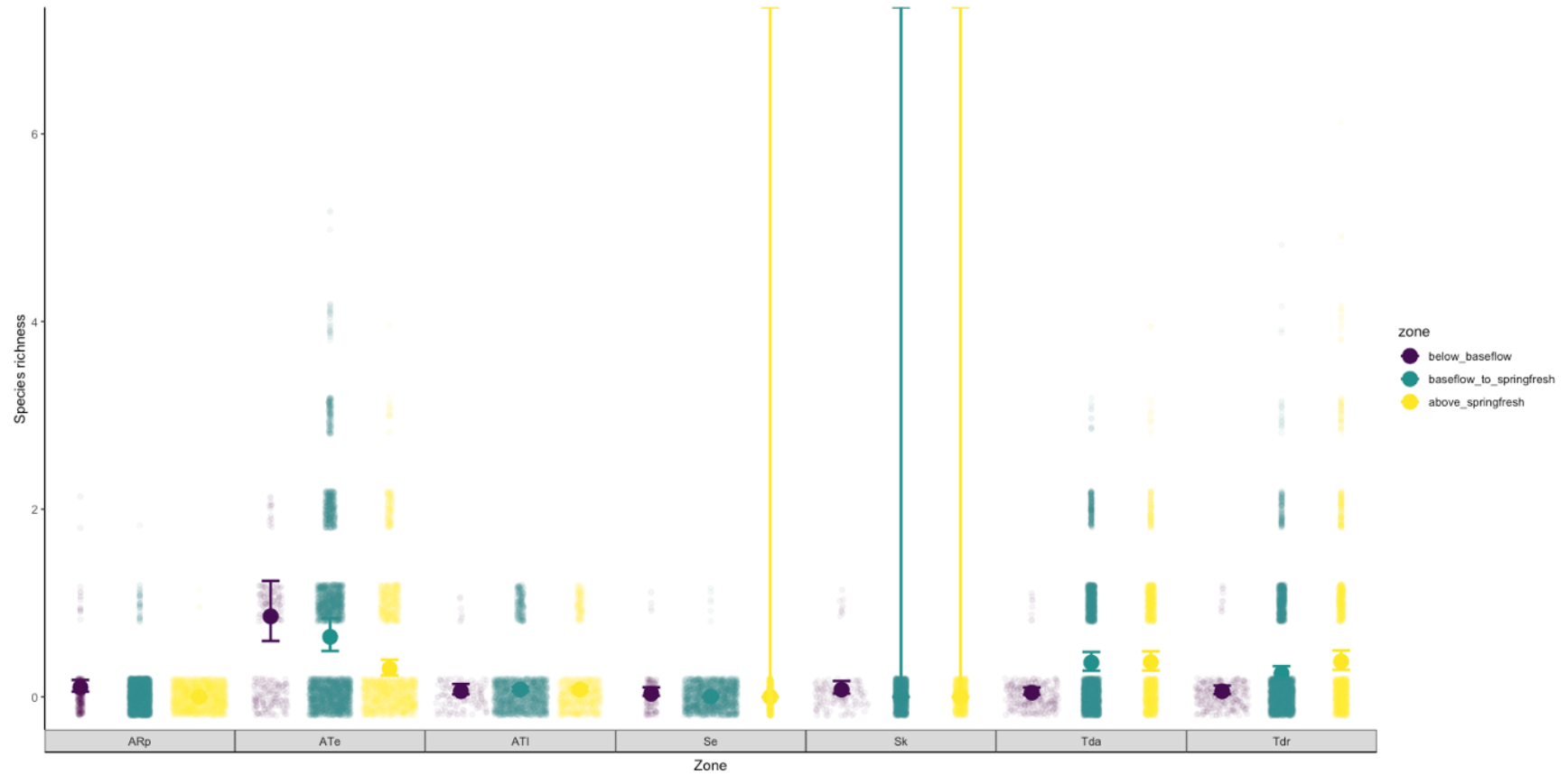


Figure 2.9: Model predicted species richness for the two way interaction between **zone** and **functional group**. Points are raw data with the spread across x-axis representing data density.

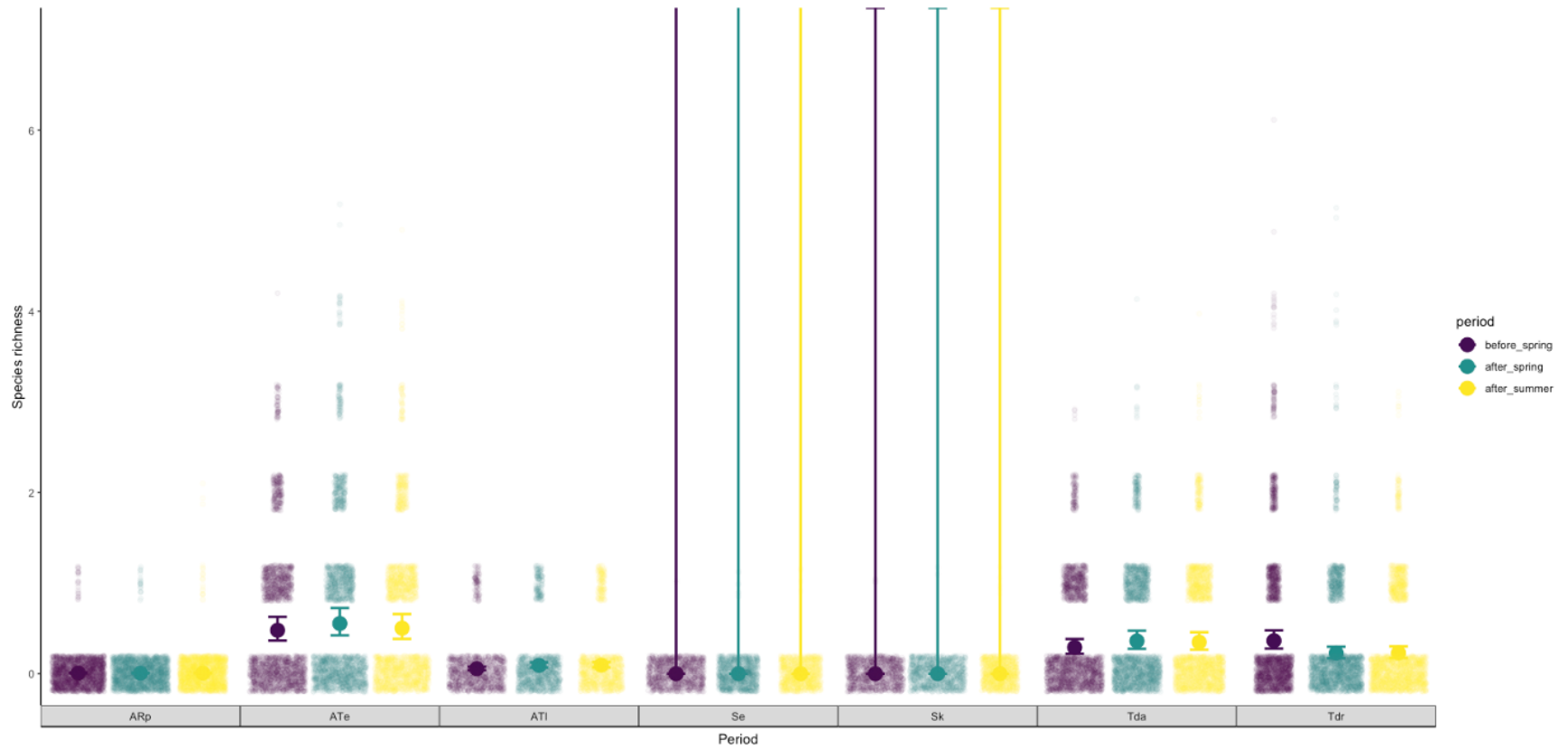


Figure 2.10: Model predicted species richness for the two way interaction between **period** and **functional group**. Points are raw data with the spread across x-axis representing data density.

2.4 Case Study Preregistration with git and GitHub

The screenshot shows a GitHub issue thread titled "Data processing for analysis #23 #23" with a "Closed" status. The issue is managed by user **csjones8**. The main content of the issue is a detailed report on data processing challenges, organized into sections: "Sites that we can't evaluate spring flow effects, or time series is limited:", "No flow elevation data (can't assess flow effects):" (listing sites like Sutherland Creek, Tullaroop Mullins, etc.), "Limited flow data (can assess effects if flow data spans survey period):" (discussing gauge data and logger data), and "No spring fresh within survey period:". The thread includes two commit updates from **csjones8** on Dec 16, 2020, and three commit updates on Dec 17, 2020, all referencing the issue.

Issue Title: Data processing for analysis #23 #23 (Closed)

Commit History:

- csjones8** added a commit that references this issue on Dec 16, 2020
Updates to species master [#23](#) e058ee2
- csjones8** on Dec 16, 2020 · edited by csjones8
Edits Collaborator Author ...
Sites that we can't evaluate spring flow effects, or time series is limited:
No flow elevation data (can't assess flow effects):
 - Sutherland Creek
 - Tullaroop Mullins
 - 12 Mile Frosts
 - Mt William Sheepwash
 - Burnt Millers
 - Moorabool Bakers Bridge**Limited flow data (can assess effects if flow data spans survey period):**
In some cases, there was no gauge data and the only data available are from the loggers installed. In some cases the logger data overlap with the veg surveys and in others they don't. None of these have data from further back in time.
 - Mt William Tobacco (data spans survey)
 - Mt William Roses Gap (data spans survey)
 - Burnt Peuckers (data spans survey)
 - MacKenzie Mt Victory (data doesn't span survey)
 - MacKenzie Tatlocks/Laharum (data doesn't span survey)
 - Glenelg Weavers (data doesn't span survey)**No spring fresh within survey period:**
 - All Glenelg River sites. Can use data for other purposes though.
All sites that can be paired with flow data have now been completed. Flow data elevations are relevant to a specific sub-transect within sites in some cases so need to be careful with data use.
- csjones8** added 3 commits that reference this issue on Dec 17, 2020
 - Updates to Glenelg Five Mile flow data and metadata [#23](#) f17c158
 - Wannon flow data correlation [#23](#) dd0eda1
 - Burnt Peuckers flow data correlation [#23](#) 5756299

Figure 2.11: Screenshot of GitHub issue thread illustrating where the preregistered analysis tasks and rationale were transparently recorded. The full thread can be viewed at https://github.com/egouldo/VEFMAP_VEG_Stage6/issues/23.



Figure 2.12: Screenshot of [comment on GitHub issue](https://github.com/egouldo/VEFMAP_VEG_Stage6/issues/71) illustrating data-dependent decision-making and where deviation from preregistration occurred during the case study. The full thread can be viewed at https://github.com/egouldo/VEFMAP_VEG_Stage6/issues/71.